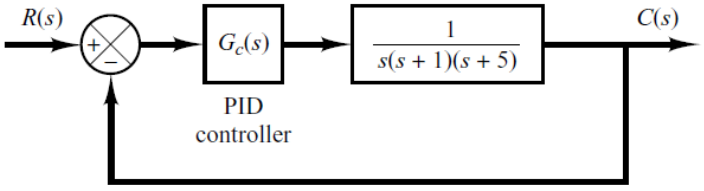
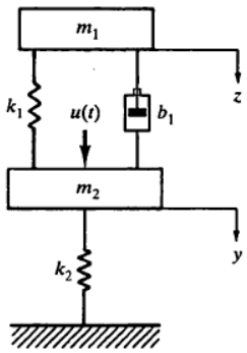
Assignment 3

1. Consider the control system shown in the figure below in which a PID controller is used to control the system.



   (a) Write the transfer function of the PID controller Gc(s)
   (b) Determine the parameters of the PID controller using Ziegler-nichols method. (Hint: use Matlab simulation to find the critical gain K_{cr} and oscillation period P_{cr} when sustained oscillations are observed.)

2. Consider the mechanical system shown in the figure:



1) Obtain a state-space representation with a state variable $\mathbf{x} = [y, \dot{y}, z, \dot{z}]^{\mathrm{T}}$ and the output $\mathbf{y} = [y, z]^{\mathrm{T}}$
2) Given $m_1 = m_2 = 1$, $k_1 = k_2 = 1$, $b_1 = 0.5$, check if the system is completely state controllable and observable.
3) Design a controller such that the system has desired poles at $-\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$, -5,-10